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CHE 416 — Past Questions & Materials
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Paper 1 of 1 — CHE 416 CHEMICAL ENGINEERING PROCESS ANALYSIS PQ

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

First Semester Examination; 2023/2024 Session

COURSE TITLE: Chemical Engineering Process Analysis

COURSE CODE: CHE 416

Instructions: Answer any four (4) questions.

Credit Units: 2

Time Allowed: 2HRS

- 1) Consider a perfectly mixed storage tank with non-isothermal conditions as shown in Figure 1 below. The liquid entered the tank with the flowrate F_{in} , density ρ_{in} and temperature T_{in} . It is heated with an external heat supply of temperature T_{st} which is assumed to be constant. The effluent stream is of flowrate F_{out} , density ρ_{out} and temperature T_{out} . Develop a mathematical model for the variation of both liquid level and temperature.

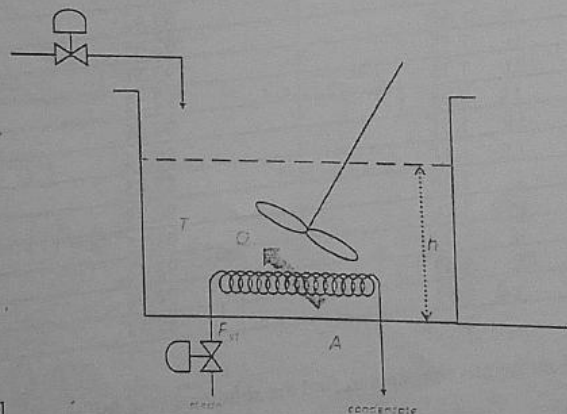


Figure 1

- 2)
- What is process modelling?
 - State three (3) importance of process modelling
 - Outline the six (6) steps involved in developing mathematical models.

- (d) State one (1) difference between:
- Theoretical model and empirical model
 - Steady state model and unsteady state model
 - Lumped parameter model and distributed parameter model
 - Classical design of experiment and statistical design of experiment.

- 3) Consider the data in Table 1 where y is the purity of oxygen produced in a chemical distillation process, and x is the percentage of hydrocarbons that are present in the main condenser of the distillation unit. Fit a simple linear regression model to the oxygen purity data in Table 1 and calculate the predicted value of the purity, y .

Table 1: Oxygen and Hydrocarbon Levels

Hydrocarbon Level $x(\%)$	Purity $y(\%)$
1 0.99	90.01
2 1.02	89.05
3 1.15	91.43
4 1.29	93.74
5 1.46	96.73
6 1.36	94.45
7 0.87	87.59
8 1.23	91.77
9 1.55	99.42
10 1.40	93.65
11 1.19	93.54
12 1.15	92.52
13 0.98	90.56
14 1.01	89.54
15 1.11	89.85
16 1.20	90.39
17 1.26	93.25
18 1.32	93.41
19 1.43	94.98
20 0.95	87.33

4. Use the Gauss-Seidel iterative process to find the solution of the following system of equations

$$4.7x_1 + 1.3x_2 - 1.6x_3 = 1.3$$

$$x_1 - 4.1x_2 + 1.1x_3 = 4.6$$

$$2.1x_1 + 1.4x_2 + 6.2x_3 = 5.2$$

CHE 416 Chemical Engineering Process Analysis: 2023/2024
1

Applying the mass balance equation.

$$[\text{Rate of mass in}] - [\text{Rate of mass out}] = [\text{Rate of mass accumulated}]$$

$$F_{in} \rho_{in} - [F_{out} \rho_{out}] = \frac{dV}{dt} \rho$$

Assume constant density

$$F_{in} - [F_{out}] = \frac{dV}{dt}$$

Dividing through by ρ .

$$F_{in} - F_{out} = \frac{dV}{dt}$$

Call $V = Ah$

$$F_{in} - F_{out} = \frac{dAh}{dt}$$

$$F_{in} - F_{out} = A \frac{dh}{dt} \quad (\text{Area is constant})$$

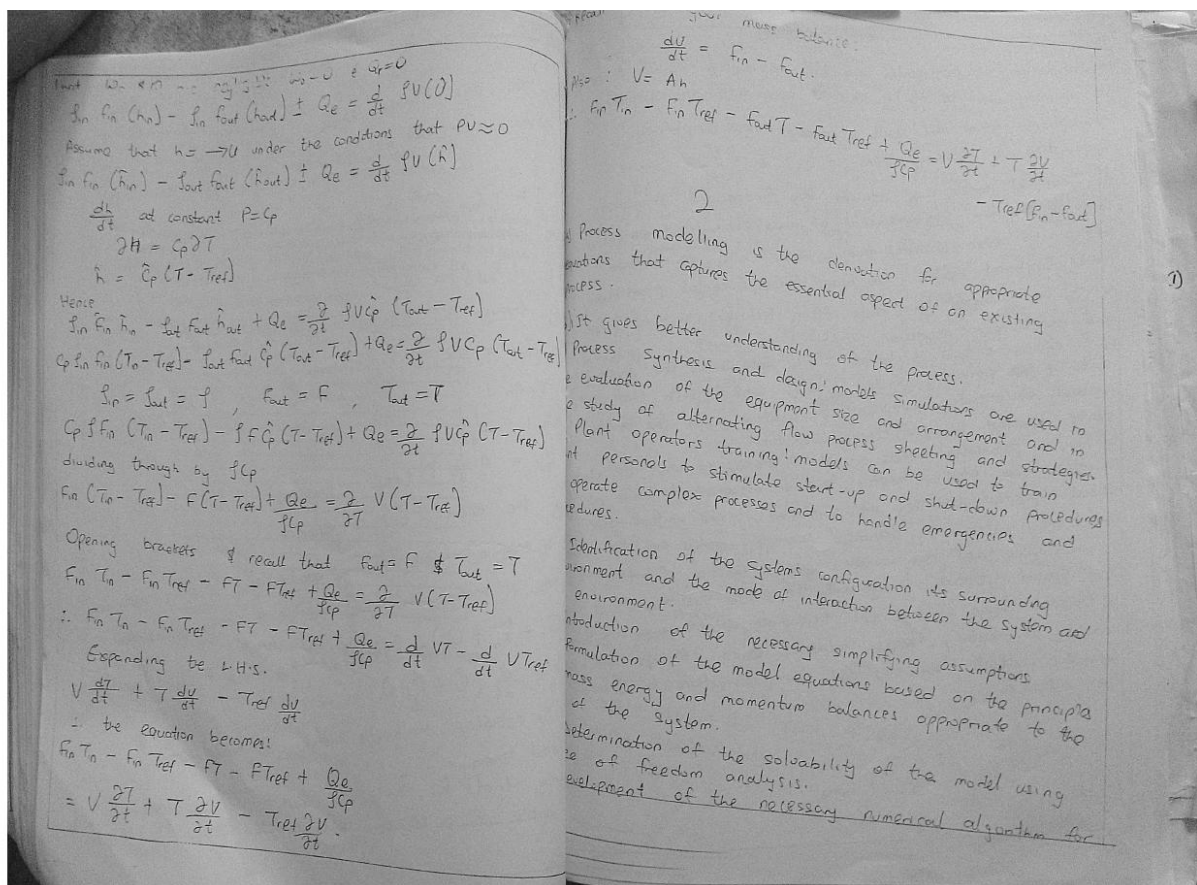
Applying the energy balance equation.

$$[\text{Rate of energy in}] - [\text{Rate of energy out}] + [\text{Rate of energy generated}] + [\text{Rate of energy exchanged}] = [\text{Rate of energy accumulated}]$$

$$F_{in}(\hat{h}_{in} + \hat{k}_{in} + \hat{Q}_{in}) - F_{out}(\hat{h}_{out} + \hat{k}_{out} + \hat{Q}_{out}) + \dot{Q}_r \pm (\dot{Q}_e - \dot{W}_o) = \frac{d}{dt}(\hat{U} + \hat{K} + \hat{Q})$$

where h = enthalpy, k = kinetic energy, Q = heat energy
 $\hat{U}$ = internal energy.

Assuming that $\hat{k}_{in}$, $\hat{k}_{out}$, $\hat{Q}_{in}$, $\hat{Q}_{out}$, $\hat{k}$, $\hat{Q}$ are assumed to be negligible for a steady state system. Also assuming



The solution of the 1^{st} situation

v) Validation of the model against experimental result to assess its validity and reliability, and to reevaluate the simplifying assumptions which may result in imposing new simplifying assumption or relaxing others.

d) Theoretical models are obtained from fundamental principles such as the law of conservation of mass, energy and momentum along with other chemical reacting kinetics and thermodynamic equilibria.

while Empirical models are based on experimental plant data, which are developed using data fitting technique such as linear and non-linear regression.

ii) A system is said to be at steady state when the physical state of the processing system remains constant with time while unsteady state process represent the situation when the process state [dependent variable] changes with time.

iii) Lumped parameter models are those models in which the state variables and other parameter have or assumed to have no spatial dependent i.e they are considered to be uniform over the entire system while distributed parameter models are the models in which state and other variables are a function of both time and spatial position through out the entire system.

iv) Classical design of experiment focuses on optimizing a single variable or factor, whereas statistical design of experiment considers multiple variables and their interactions to optimize the outcome.

X	Y	XY	X ²
0.99	90.01	89.1099	87.3366
1.02	89.05	90.831	1.0404
1.15	91.43	105.1445	1.3225
1.29	93.74	120.9246	1.6641
1.46	96.73	141.2258	2.1316
1.36	94.45	128.452	1.8496
0.87	87.59	76.2033	0.7569
1.23	91.77	112.8771	1.5129
1.53	99.42	154.101	2.4025
1.40	93.65	131.11	2.7440
1.19	93.54	111.3126	1.4161
1.15	92.52	106.398	1.3225
0.98	90.56	88.7488	0.9604
1.01	89.54	90.4354	1.0201
1.11	89.85	99.7335	1.2321
1.20	90.39	108.468	1.4400
1.26	93.25	117.495	1.5876
1.32	93.41	123.3012	1.7424
1.43	94.98	135.8214	2.0449
0.95	87.33	82.9635	0.9025
23.92	1843.21	2214.6566	29.2892

$$\sum Y = 1843.21$$

$$\sum XY = 2214.6566$$

$$\sum X^2 = 29.2892$$

where $n = 20$

$$1843.21 = 20a + 23.92b$$

$$2214.6566 = 23.92a + 29.2892b$$

Solving simultaneously

$$a = 74.2833 \quad b = 14.9475$$

5a) Use the fourth order Runge-Kutta algorithm with a step size $h = 0.5$ to solve the initial value problem

$$y'' = (x + 2)y - 2y', \text{ with } x=0, y = 0, y' = 0 \text{ and } x=0(0.5)1.5.$$

5b) Find the solution to the equation below using numerical approximation approach

$$4 \frac{\partial f(x,y)}{\partial x} + 2 \frac{\partial f(x,y)}{\partial y} = 3, \text{ for } 0 \leq x \leq 1 \text{ and for } 0 \leq y \leq 1 \text{ given that the boundary conditions are } f(x, 0) = f(x, 1) = f(0, x) = 10 \text{ and } \frac{\partial f(x,y)}{\partial x} = 2. \text{ Use a mesh of size } 1/3 \text{ in both the } x \text{ and the } y \text{ direction.}$$

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, E.T.U.U.



COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

First Semester Examination; 2022/2023 Session

COURSE TITLE: CHEMICAL ENGINEERING PROCESS ANALYSIS

COURSE CODE: CHE 416

INSTRUCTIONS: Answer ANY four (4) questions.

Credit Units: 2

Time Allowed: 2 hrs.

- Describe the term 'Mathematical Model'
 - Use the Gauss-Seidel iterative process to find the solution of the following system of equations

$$\begin{aligned} 4.7x_1 + 1.3x_2 - 1.6x_3 &= 1.3 \\ x_1 - 4.1x_2 + 1.1x_3 &= 4.6 \\ 2.1x_1 + 1.4x_2 + 6.2x_3 &= 5.2 \end{aligned}$$

a_{11}	a_{12}	a_{13}	b_1
a_{21}	a_{22}	a_{23}	b_2
a_{31}	a_{32}	a_{33}	b_3
- An isothermal PFR has the reaction $A \rightarrow P$ taking place in it. Assuming that the density, concentration and velocity change with the axial direction only, develop a model that describes the reaction process in the tube.
 - Briefly explain the following
 - Lumped parameter model
 - Empirical model
 - Distributed parameter model
 - Analytical model
- Liquid with density ρ_{in} and temperature T_{in} (K) flows at a rate of F_1 (m³/s) into a perfectly mixed storage tank under non-isothermal conditions. It is heated with the aid of an external heat supply with temperature T_{H1} which is assumed constant. The effluent stream is of flow rate F_{out} (m³/s) and density ρ_{out} and temperature T . With above description, develop a mathematical model for both the variation of liquid level and its temperature.
- A chemical reaction takes place in a series of four continuous stirred tank reactors arranged in figure 1. The chemical reaction is a first-order irreversible reaction of the type $A \xrightarrow{k_i} B$. The conditions of temperature in each reactor are such that the value of the rate constant k_i is different in each reactor. Also, the volume of each reactor V_i

is different. The values of k , and v , are given in Table Q3. The following assumptions were made regarding this system:

- The system is at steady state.
- The reactions are in the liquid phase.
- There is no change in volume or density of the liquid.
- The rate of disappearance of component A in each reactor is given by $R_i = V_i k_i C_A$, mol/h

- Set up the material balance (model) equation for each of the four reactors.
- Use the Gauss Seidel method to solve the model equation

Table Q4

Reactor	v_i (litre)	k_i (per hour)
1	1000	0.1
2	1500	0.2
3	100	0.3
4	500	0.4

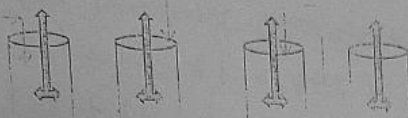


Figure Q4

- A perfectly mixed tank (CSTR) has a liquid phase, first order, irreversible chemical reaction expressed as $A \xrightarrow{k_1} B$ taking place. The feed enters the reactor with a volumetric flow rate F_{in} , density, δ_{in} (kg/m^3) and concentration C_{Ain} (mol/m^3). The output comes out of the reactor at a volumetric flow rate F_{out} , density, δ_{out} (kg/m^3), and concentration, C_{Aout} (mol/m^3). Assuming a constant density, volume, non-isothermal and exothermic conditions where the heat generated in the reactor is removed via a cooling system, develop a model for the variation of specie A.
- An isothermal constant volume CSTR in which a liquid phase chemical reaction $A \xrightarrow{k} B$ has the unsteady state model given as $V \frac{dC_A}{dt} = F C_{Ain} - F C_A - (-r_A)V$.
Using the Newton Raphson's method for a range of 0-5 and an initial root of 1.0, find the concentration of A at the exit stream at steady state if the reaction is in the third order with the following assumptions: $f=10\text{m}^3/\text{s}$, $k = 2\text{mol}^{-2}\text{m}^2\text{s}^{-1}$, $v=25\text{m}^3$, $C_{Ain} = 0.1\text{mol/m}^3$.

CHE 416 2022/2023

(1)

- mathematical model is a mathematical expression that describes a process, it could be analytical or empirical. The model captures the essential aspects of an existing system.

Given the system equations

$$7x_1 + 1.3x_2 - 1.6x_3 = 1.3$$

$$-4.1x_2 + 1.1x_3 = 4.6$$

$$x_1 + 1.4x_2 + 6.2x_3 = 5.2$$

∴ Test for Diagonal Dominance

$$1 > |1.3| + |-1.6| \quad \text{YES}$$

$$1 > |1| + |1.1| \quad \text{YES}$$

$$2 > |2.1| + |1.4| \quad \text{YES}$$

∴ It is diagonally dominant

arrangement and denotations

$$1 \quad a_{12} \quad a_{13} \quad b_1$$

$$1 \quad a_{22} \quad a_{23} \quad b_2$$

$$1 \quad a_{32} \quad a_{33} \quad b_3$$

using

$$= \frac{1}{a_{11}} (b_1 - a_{12}x_2(r-1) - a_{13}x_3(r-1))$$

$$= \frac{1}{a_{22}} (b_2 - a_{21}x_1(r) - a_{23}x_3(r-1))$$

$$= \frac{1}{a_{33}} (b_3 - a_{31}x_1(r) - a_{32}x_2(r))$$

$x_1^{(1)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(0)} + 1.6x_3^{(0)})$
 $x_1^{(1)} = \frac{1}{4.7} (1.3 - 1.3(0) + 1.6(0))$
 $x_1^{(1)} = 0.3404$
 $x_2^{(1)} = \frac{1}{-4.1} (4.6 - 1x_1^{(1)} - 1.1x_3^{(0)})$
 $x_2^{(1)} = \frac{1}{-4.1} (4.6 - 1(0.3404) - 1.1(0))$
 $x_2^{(1)} = -0.7706$
 $x_3^{(1)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(1)} - 1.4x_2^{(1)})$
 $x_3^{(1)} = \frac{1}{6.2} (5.2 - 2.1(0.3404) - 1.4(-0.7706))$
 $x_3^{(1)} = 0.8974$
RESULT FROM FIRST ITERATION
 $x_1^{(1)} = 0.3404; x_2^{(1)} = -0.7706; x_3^{(1)} = 0.8974$
WHEN $r=2$ (2ND ITERATION)
 $x_1^{(2)} = 0.3404; x_2^{(2)} = -0.7706; x_3^{(2)} = 0.8974$
 $x_1^{(2)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(1)} + 1.6x_3^{(1)})$
 $x_1^{(2)} = \frac{1}{4.7} (1.3 - 1.3(-0.7706) + 1.6(0.8974))$
 $x_1^{(2)} = 0.7939$
 $x_2^{(2)} = \frac{1}{-4.1} (4.6 - 1x_1^{(2)} - 1.1x_3^{(1)})$
 $x_2^{(2)} = \frac{1}{-4.1} (4.6 - 1(0.7939) - 1.1(0.8974))$
 $x_2^{(2)} = -0.6876$
 $x_3^{(2)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(2)} - 1.4x_2^{(2)})$
 $x_3^{(2)} = \frac{1}{6.2} (5.2 - 2.1(0.7939) - 1.4(-0.6876))$
 $x_3^{(2)} = 0.7251$
RESULT FROM 2ND ITERATION
 $x_1^{(2)} = 0.7939; x_2^{(2)} = -0.6876; x_3^{(2)} = 0.7251$
3RD ITERATION (WHEN $r=3$)
 $x_1^{(3)} = 0.7939; x_2^{(3)} = -0.6876; x_3^{(3)} = 0.7251$
 $x_1^{(3)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(2)} + 1.6x_3^{(2)})$
 $x_1^{(3)} = \frac{1}{4.7} (1.3 - 1.3(-0.6876) + 1.6(0.7251))$
 $x_1^{(3)} = 0.7136$
 $x_2^{(3)} = \frac{1}{-4.1} (4.6 - 1x_1^{(3)} - 1.1x_3^{(2)})$
 $x_2^{(3)} = \frac{1}{-4.1} (4.6 - 1(0.7136) - 1.1(0.7251))$
 $x_2^{(3)} = -0.5879$
 $x_3^{(3)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(3)} - 1.4x_2^{(3)})$
 $x_3^{(3)} = \frac{1}{6.2} (5.2 - 2.1(0.7136) - 1.4(-0.5879))$
 $x_3^{(3)} = -0.7534$

$x_1^{(3)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(2)} - 1.4x_2^{(2)})$
 $x_2^{(3)} = \frac{1}{6.2} (5.2 - 2.1(0.736) - 1.4(-0.7534))$
 $x_3^{(3)} = 0.7671$
 Repeat until it converges
 $x_1^{(4)} = 0.736; x_2^{(4)} = -0.7534; x_3^{(4)} = 0.7671$
 4th iteration value is 4
 $x_1^{(5)} = 0.736, x_2^{(5)} = -0.7534, x_3^{(5)} = 0.7671$
 $x_1^{(4)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(3)} + 1.6x_3^{(3)})$
 $x_1^{(4)} = \frac{1}{4.7} (1.3 - 1.3(-0.7534) + 1.6(0.7671))$
 $x_1^{(4)} = 0.7461$
 $x_2^{(4)} = \frac{1}{-4.1} (4.6 - 1x_1^{(4)} - 1.1x_3^{(4)})$
 $x_2^{(4)} = \frac{1}{-4.1} (4.6 - 1(0.7461) - 1.1(0.7671))$
 $x_2^{(4)} = -0.7342$
 $x_3^{(4)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(4)} - 1.4x_2^{(4)})$
 $x_3^{(4)} = \frac{1}{6.2} (5.2 - 2.1(0.7461) - 1.4(-0.7342))$
 $x_3^{(4)} = 0.7518$

$x_1^{(5)} = 0.736, x_2^{(5)} = -0.7534, x_3^{(5)} = 0.7671$
 $x_1^{(5)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(4)} + 1.6x_3^{(4)})$
 $x_1^{(5)} = \frac{1}{4.7} (1.3 - 1.3(-0.7342) + 1.6(0.7518))$
 $x_1^{(5)} = 0.7356$
 $x_2^{(5)} = \frac{1}{-4.1} (4.6 - 1x_1^{(5)} - 1.1x_3^{(5)})$
 $x_2^{(5)} = \frac{1}{-4.1} (4.6 - 1(0.7356) - 1.1(0.7518))$
 $x_2^{(5)} = -0.7408$
 $x_3^{(5)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(5)} - 1.4x_2^{(5)})$
 $x_3^{(5)} = \frac{1}{6.2} (5.2 - 2.1(0.7356) - 1.4(-0.7408))$
 $x_3^{(5)} = 0.7568$
 6th iteration value is 6
 $x_1^{(6)} = 0.7356, x_2^{(6)} = -0.7408, x_3^{(6)} = 0.7568$
 $x_1^{(6)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(5)} + 1.6x_3^{(5)})$
 $x_1^{(6)} = \frac{1}{4.7} (1.3 - 1.3(-0.7408) + 1.6(0.7568))$
 $x_1^{(6)} = 0.7391$
 $x_2^{(6)} = \frac{1}{-4.1} (4.6 - 1x_1^{(6)} - 1.1x_3^{(6)})$

$$= 1000000 \cdot (1 + 0.05)^{10} - 1000000 = 628894.62$$

$$t_2(6) = -0.7386$$

$$f^{(6)} = \frac{1}{6 \cdot 2} [5 \cdot 2 - 21 + 1^{(6)} - 1 \cdot 4 + 2^{(6)}]$$

$$x_3(6) = \frac{1}{6-2} [5.2 - 2.1(0.7391) - 1.4(-0.7386)]$$

$$13^{(6)} = 0.7552$$

FHA Regulation with R2

$$x^{(6)} = 0.7391, x_2^{(6)} = 0.7386, x_3^{(6)} = 0.7552$$

$$\Rightarrow x_1(7) = \frac{1}{4-7} (1-3 - 1-3 + 2(6) + 1-6 + 3(6))$$

$$x^{(7)} = \frac{1}{4.7} [(3 - 1.3(-0.7386) + 1.6(0.7552))] =$$

$$4^{(7)} = 20.7380$$

$$x_2^{(7)} = \frac{1}{-41} (46 - 1x_1^{(7)} - 1.1x_3^{(6)})$$

$$x_2(4) = \frac{1}{-41} (46 - 1(0.7380) - 1(0.7552))$$

$$x_2^{(7)} = -0.7393$$

$$x_3^{(7)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(7)} - 1.4x_2^{(7)})$$

$$x_3^{(7)} = \frac{1}{6.2} [5.2 - 2.1(0.7380) - 1.4(-0.7393)]$$

$$L_3(7) = 0.7557$$

48: We could actually stop at this point, but the goal is to get to a point, where the values will be repeating themselves (i.e. at the 0th iteration)

$$+ (7) = 0.7380; \quad \frac{1}{2} (7) =$$

87A ITERATION (11)

$-7380, y_2(7) = -0.7380$

$$f'(8) = \frac{1}{47} (1.3 - 1.3 \times 2) \quad (7) \quad 20.7557$$

$$+^{(8)} = \frac{1}{47} ((13 - 13(-0.7393)) + 16(0.7557))$$

4 (8) 20.7383

$$x_2^{(5)} = \frac{1}{-4-1} (4-5 - 1 \cdot x_1^{(5)} - (-1) \cdot x_3^{(7)})$$

$$f_2(f) = \frac{1}{-4.1} (46 - 1(0.7383) - 1(0.7557))$$

$$x_2(\varphi) = -0.7391$$

$$z(s) = \frac{1}{6.2} (s-2 - 2.1x_1(s) - (-4x_2(s)))$$

$$f_2(5) = \frac{1}{6.2} (5.2 - 2.1(0.7383) - 1.4(-0.7391))$$

$$x_3(8) = 0.7585$$

Almost true!!!

9TH 1724714 (124714) r=9
(P) 20-738 (A)

$$x_1^{(9)} = \frac{1}{13 - 1.3 \cdot (9)} \quad (x_2^{(9)} = -0.7391, x_3^{(9)} = 0.7555)$$

$$x(q) = \frac{1}{4-7} (13 - 13x_2(q) + 1.6x_3(q))$$

$$x_1(q) = \frac{1}{47} (13 - 13(-0.7391) + 16(0.7555))$$

$x_1(q) = 0.7382$

$$x_1(9) = 0.7382$$

$x_1^{(9)} = \frac{1}{-4.1} (4.6 - 1.4x_2^{(9)} - 1.4x_3^{(9)})$
 $x_2^{(9)} = \frac{1}{-4.1} (4.6 - 1(0.7382) - 1(0.7556))$
 $x_3^{(9)} = -0.7392$
 $x_3^{(9)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(9)} - 1.4x_2^{(9)})$
 $x_3^{(9)} = \frac{1}{6.2} (5.2 - 2.1(0.7382) - 1.4(-0.7392))$
 $x_3^{(9)} = 0.7556$
 Finally!!
 RESULTS FROM 9TH ITERATION
 $x_1^{(9)} = 0.7382, x_2^{(9)} = -0.7392, x_3^{(9)} = 0.7556$
 10TH ITERATION WITH $r=0$
 $x_1^{(10)} = 0.7382, x_2^{(10)} = -0.7392, x_3^{(10)} = 0.7556$
 $x_1^{(10)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(10)} + 1.6x_3^{(10)})$
 $x_1^{(10)} = \frac{1}{4.7} (1.3 - 1.3(-0.7392) + 1.6(0.7556))$
 $x_1^{(10)} = 0.7383$
 $x_2^{(10)} = \frac{1}{-4.1} (4.6 - 1.4x_1^{(10)} - 1.4x_3^{(10)})$
 $x_2^{(10)} = \frac{1}{-4.1} (4.6 - 1(0.7383) - 1(0.7556))$
 $x_2^{(10)} = -0.7392$
 $x_3^{(10)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(10)} - 1.4x_2^{(10)})$
 $x_3^{(10)} = \frac{1}{6.2} (5.2 - 2.1(0.7383) - 1.4(-0.7392))$

$x_3^{(10)} = 0.7556$
 observe carefully the trend from the 7th - 10th iteration, how the values starts becoming similar and unchanging, further iterations will then give repeated values. Hence iteration ends here.

	0	1	2	3	4	5	6	7	8	9	10
x_1	1	0.3404	0.7939	0.7336	0.7461	0.7356	0.7397	0.7382	0.7383	0.7382	0.7383
x_2	1	-0.7206	-0.6266	-0.7334	-0.7342	-0.7458	-0.7386	-0.7373	-0.7376	-0.7376	-0.7376
x_3	1	0.8974	0.7251	0.7691	0.7516	0.7568	0.7552	0.7557	0.7557	0.7556	0.7556

$X = \begin{pmatrix} 0.7383 \\ -0.7392 \\ 0.7556 \end{pmatrix}$

(2)
 Con $\int_{z_1}^{z_2} v(z) dz$

 MODEL FORMULATION
 This is a PFR model, and in modelling a PFR, begin with a macroscopic component balance for space A around the differential element of width δz and constant cross-sectional area S .
 Hence, the plug flow is being modelled, bearing in mind that mass transfer in the differential element occurs by two mechanisms: convection and diffusion.

Rate of moles of A
Rate of moles of B
Rate of accumulation of moles of A
Rate of accumulation of moles of B

Rate of consumption of moles of species A by reaction within the differential element
Rate of accumulation of species A
Rate of accumulation of species B

HB: The generation flow of the flow that is associated with steady and occurs in bulk

$$\left(C_A V_1 + H_A S_1 \right) - \left(C_A V_2 + H_A S_2 \right) - (C_A) S \delta z = \frac{\partial}{\partial t} (C_A S \delta z)$$

HB: PFR models are actually (distributed) parameter models. Hence the equation obtained will be in a partial differential equation format!

Assume first order: $-r_A = k C_A$

$$C_A V_1 + H_A S_1 - (C_A V_2 + H_A S_2 + \delta C_A V_2 + \delta H_A S_2) - k C_A S \delta z = \frac{\partial}{\partial t} (C_A S \delta z) \quad (1)$$

$$C_A V_1 + H_A S_1 - C_A V_2 - H_A S_2 - \delta C_A V_2 - \delta H_A S_2 - k C_A S \delta z = \frac{\partial}{\partial t} (C_A S \delta z) \quad (2)$$

$$- \delta C_A V_2 - \delta H_A S_2 - k C_A S \delta z = \frac{\partial}{\partial t} (C_A S \delta z) \quad (3)$$

Dividing through by $S \delta z$

$$\frac{-\delta C_A V_2}{S \delta z} - \frac{\delta H_A S_2}{S \delta z} - \frac{k C_A S \delta z}{S \delta z} = \frac{\partial}{\partial t} (C_A S \delta z) / S \delta z$$

$$\frac{-\delta C_A V_2}{S \delta z} - \frac{\delta H_A S_2}{S \delta z} - k C_A = \frac{\partial}{\partial t} C_A \quad (4)$$

HB: $\delta z \rightarrow 0, \delta t \rightarrow 0$

$$\frac{\partial C_A V}{\partial z} - \frac{\partial H_A S}{\partial z} - k C_A = \frac{\partial C_A}{\partial t} \quad (5)$$

HB: PFR is like a tank

$$H_A = -V_A \frac{\partial C_A}{\partial z}$$

$$-V_A \frac{\partial C_A}{\partial z} - C_A \frac{\partial V}{\partial z} - \frac{\partial}{\partial z} \left(-V_A \frac{\partial C_A}{\partial z} \right) - k C_A = \frac{\partial C_A}{\partial t} \quad (6)$$

If the fluid is actually incompressible, the velocity does not change with position

$$\therefore \frac{\partial V}{\partial z} = 0 \quad (\text{Continuity equation})$$

$$-V \frac{\partial C_A}{\partial z} + V_A \frac{\partial C_A}{\partial z} - k C_A = \frac{\partial C_A}{\partial t} \quad (7)$$

At steady state, $\frac{\partial C_A}{\partial t} = 0$, Accumulation = 0

$$-V \frac{\partial C_A}{\partial z} + V_A \frac{\partial C_A}{\partial z} - k C_A = 0 \quad (8)$$

$$V_A \frac{\partial C_A}{\partial z} = k C_A + V \frac{\partial C_A}{\partial z}$$

$$\therefore \frac{\partial C_A}{\partial z} = \left(\frac{k}{V_A} \right) C_A + \left(\frac{V}{V_A} \right) \frac{\partial C_A}{\partial z}$$

$$\frac{\partial C_A}{\partial z} - \left(\frac{V}{V_A} \right) \frac{\partial C_A}{\partial z} = \left(\frac{k}{V_A} \right) C_A = 0$$

- models with an ODE (ordinary differential equation)

$$= \frac{d^2 C_A}{dz^2} - \left(\frac{V}{\delta z} \right) \frac{dC_A}{dz} - \left(\frac{k_c}{\delta z} \right) C_A = 0 \quad (10)$$

The ODE shown above, can be solved analytically using the following boundary condition:

At $z=0$, $C_A(0) = C_{A0}$ which gives the concentration of A at the reactant for entrance.

At $z=L$, $\frac{dC_A}{dz} = 0$ which means, there is no flux at the exit length of the reactor.

2B)

(i) Lumped parameter model

These are models in which state variables and other parameters have or assume to have no spatial dependence (they are considered to be uniform over an entire system). In this case, the time for unsteady state model is the only independent variable. A good example includes a perfectly mixed CSTR, distillation column etc. They are solved by carrying out 'macroscopic' balances for the process.

(ii) Empirical model

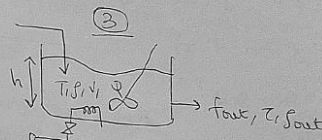
Our analysis is called a statistical / probabilistic / non-deterministic model, it is a set of distributions that are regarded as assumptions, that generated some discrete data, in other words, it is a type of mathematical model that compares of assumptions undertaken to derive the data generated process.

3. DIFFERENTIAL PARAMETER MODEL

These are models in which the state and other variables are a function of both time and space (spatial position), in this case, modelling takes into account the variation of the variables with time and space. Examples could include PFR, Heat exchanger and packed column. These models are usually obtained by writing 'macroscopic balances' for the process.

(i) Analytical model

Our trials called 'Theoretical models' are 'first principle' fundamental models. These are models obtained by first principles such as the 'law of conservation of mass' and 'thermodynamics' along with other chemical relations. They are however, generally difficult to obtain and sometimes hard to solve.



System characterization

The system is viewed as the liquid in the tank having non-isothermal conditions, and it is perfectly mixed, the liquid enters the tank and takes liquid out of the tank as well as interacting with the surrounding. The liquid is being heated with a constant temperature from an external heat supply.

Assumptions

- 1) The system is perfectly mixed, hence the density of the effluent stream (output) is the same as that of the input.
- 2) The system (tank) is operating under 'non-steady-state' conditions.

Model Formulation

Mass Balance

$$\left[\text{Rate of mass in} \right] - \left[\text{Rate of mass out} \right] = \left[\text{Rate of mass accumulation} \right]$$

$$\rho_{in} \dot{V}_{in} - \rho_{out} \dot{V}_{out} = \frac{d(\rho V)}{dt} \quad (1)$$

From assumption 1, since the density remains constant

$$\therefore \dot{V}_{in} = \dot{V}_{out} = \dot{V} \quad (\text{assumed})$$

$$\therefore \frac{dV}{dt} = \dot{V}_{in} - \dot{V}_{out} \quad (2)$$

$$\frac{dV}{dt} = \dot{V}_{in} - \dot{V}_{out}$$

$$\frac{dV}{dt} = \dot{V}_{in} - \dot{V}_{out} \quad (3)$$

Recall: The mathematical model is to develop for both the variation of liquid level (height h) and temperature T .

Since $V = Ah$

$$\therefore \frac{d(Ah)}{dt} = \dot{V}_{in} - \dot{V}_{out} \quad (4)$$

eqn (4) is a mass balance equation for the system.

DEGREE OF FREEDOM ANALYSIS

parameter of constant value = ρ, ρ

number of unknown variables = $h, \dot{V}_{out}$

$$\therefore M_v = 2$$

number of equations (M_e) = 1

Degree of Freedom $M_d = M_v - M_e$

$$= 2 - 1 = 1$$

When the degree of freedom is 1, the system is not exactly specified, but when it is 0, it is exactly specified. For the system to be exactly specified, we need one more equation.

extra equation relates output flowrate ($\dot{V}_{out}$) to the liquid level by

$$\dot{V}_{out} = \alpha \sqrt{h}$$

where α = discharge coefficient

we make equation for the variation of liquid level

$$\frac{d(Ah)}{dt} + \dot{V}_{out} = \dot{V}_{in} \quad (5)$$

where $\dot{V}_{out} = \alpha \sqrt{h}$

Energy Balance

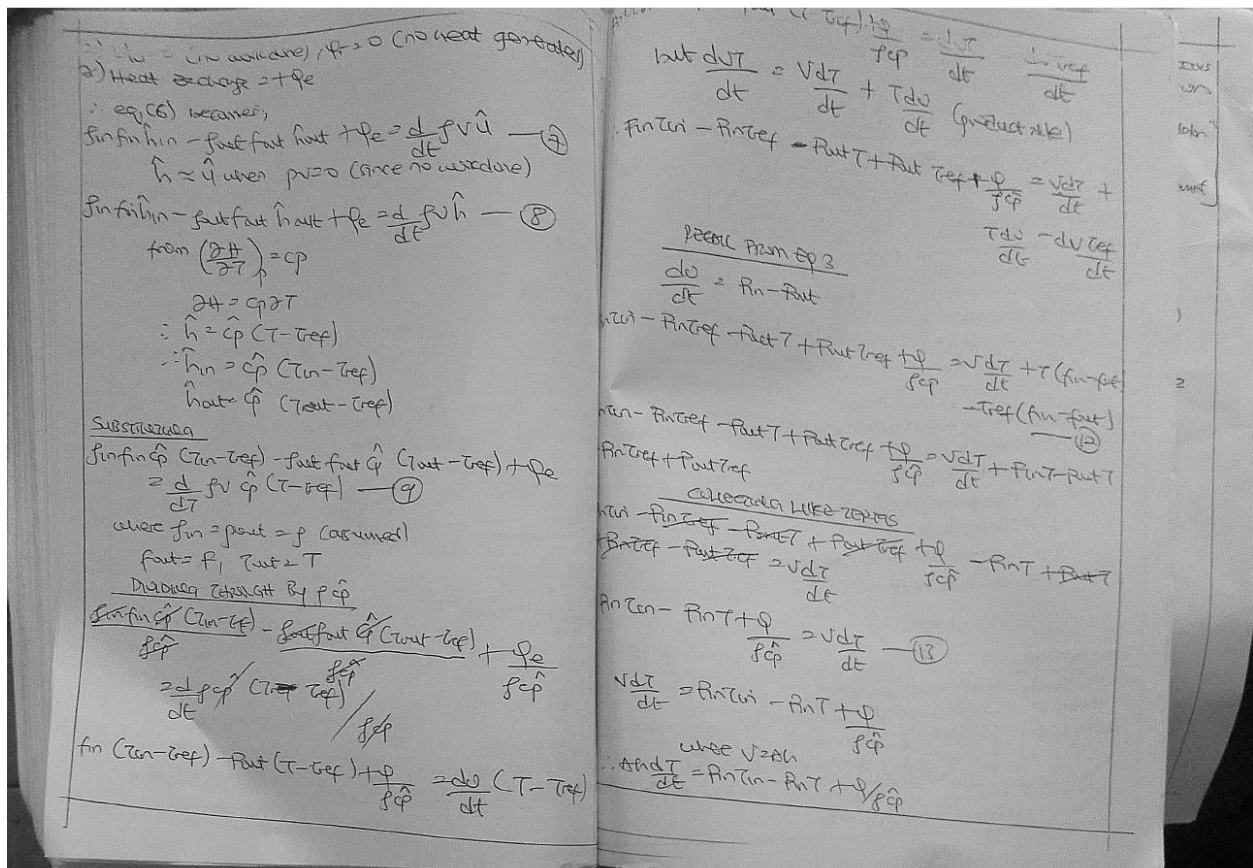
$$\left[\text{Rate of energy in} \right] - \left[\text{Rate of energy out} \right] + \left[\text{Rate of energy generated} \right] + \left[\text{Rate of energy consumed} \right] = \left[\text{Rate of energy accumulation} \right]$$

$$\dot{Q}_{in} (\dot{m}_{in} \hat{h}_{in} + \dot{V}_{in} \hat{h}_{in}) - \dot{Q}_{out} (\dot{m}_{out} \hat{h}_{out} + \dot{V}_{out} \hat{h}_{out}) + \dot{Q}_{gen} - \dot{Q}_{cons} = \frac{d(\dot{Q}_{acc})}{dt}$$

$$\dot{Q}_{in} \pm (\dot{Q}_{gen} - \dot{Q}_{cons}) = \frac{d(\dot{Q}_{acc})}{dt} \quad (6)$$

Assumptions

$\dot{Q}_{in}, \dot{Q}_{out}, \dot{V}_{in}, \dot{V}_{out}, \hat{h}_{in}, \hat{h}_{out}$ are negligible (since energy and potential energy)



Steady state equation

$$\frac{dT}{dt} = \frac{F_{in}T_{in}}{\rho V} - \frac{F_{out}T}{\rho V} + \frac{Q}{\rho V}$$

$$\frac{dT}{dt} + \frac{F_{in}T}{\rho V} = \frac{F_{out}T}{\rho V} + \frac{Q}{\rho V} \quad (14)$$

Degree of Freedom

parameter of constant = C_p, A, F
 Fixed variable = F_{in}, T_{in}
 remaining unknown = T, h, F_{out}, Q_e
 $\therefore$ No. of variable = 4
 $\therefore$ No. of equation = 3
 $\therefore$ $4 - 3 = 1$
 $\therefore$ $14 - 3 = 11$

Hence the fifth equation which will make the problem exactly specified is given as

$$Q_e = UA(T_s - T)$$

where U = Heat transfer coefficient
 A = Total Area
 T_s = Temperature of the source
 T = Temperature of the water

(4)

Assumption

It is a steady-state reaction
 $\therefore$ there is no accumulation, $\frac{dC}{dt} = 0$

$$\left[\text{mole of } A \right]_{in} - \left[\text{mole of } A \right]_{out} = \left[\text{mole of } A \right]_{consumed} = \frac{dC_A}{dt}$$

For CSTR 1

$$V_1 C_{A0} - V_1 C_{A1} - (F_{A1} V_1) = 0$$

For CSTR 2

$$\text{input} - \text{output} - \left[\text{mole of } A \text{ consumed} \right] = 0$$

$$V_1 C_{A1} + V_2 C_{A2} - (V_1 + V_2) C_{A2} - (F_{A2} V_2) = 0$$

where $F_{A2} = k_2 C_{A2}$

$$V_1 C_{A1} + V_2 C_{A2} - (V_1 + V_2) C_{A2} - (0.2 \times C_{A2} \times 1500) = 0$$

$$1500 C_{A1} + 1500 C_{A2} - 1500 C_{A2} - 300 C_{A2} = 0$$

$$1500 C_{A1} + 1500 C_{A2} - 1500 C_{A2} = 0$$

$$1500 C_{A1} + 1500 C_{A2} = 1500 C_{A2} \quad (12)$$

For CSTR 3

$$\text{input} - \text{output} - \left[\text{mole of } A \text{ consumed} \right] = 0$$

$$V_3 C_{A2} + V_4 C_{A4} - (V_3 + V_4) C_{A4} - (F_{A4} V_4) = 0$$

where $F_{A4} = k_4 C_{A4}$

$$V_3 C_{A2} + V_4 C_{A4} - 1200 C_{A4} - 300 C_{A4} = 0$$

$$1100 C_{A2} + 1200 C_{A4} = 1500 C_{A4} \quad (13)$$

For CSTR 4

$$\text{input} - \text{output} - \left[\text{mole of } A \text{ consumed} \right] = 0$$

$$V_4 C_{A4} - V_4 C_{A5} - (F_{A5} V_5) = 0$$

$1100 C_2 - 1100 C_4 = 14400$
 $1100 C_2 - 1100 C_4 = 14400 \Rightarrow 14400 C_4 = 0$
 $1100 C_2 - 1100 C_4 = 2200 C_4 = 0$
 $\therefore 1100 C_2 - 1320 C_4 = 0 \quad (4)$

Analysis of Equations

$C_1 - C_4 = 0.1 C_4 \quad (1)$
 $1000 C_1 + 100 C_2 = 1400 C_2 \quad (2)$
 $1000 C_2 + 100 C_4 = 1230 C_2 \quad (3)$
 $1100 C_2 = 1320 C_4 \quad (4)$

From eq (1)

$C_1 - C_4 = 0.1 C_4$
 $C_1 = 1.1 C_4$
 $\therefore C_1 = \frac{1}{1.1} C_4$

Assume that C_4 is known to be 1 mol/litre

$\therefore C_1 = 0.9091 \text{ mol/litre}$

which reduces the equations to three.

$\therefore$ from eq (2)
 $1000 C_1 + 100 C_2 = 1400 C_2$
 $1000 (0.9091) + 100 C_2 = 1400 C_2$
 $909.1 + 100 C_2 = 1400 C_2$

Rearranging the equations

$1400 C_2 - 100 C_2 = 909.1 \quad (1)$
 $1100 C_2 - 1230 C_2 = 100 C_4 = 0 \quad (2)$
 $1100 C_2 - 1320 C_4 = 0 \quad (3)$

using Gauss-Seidel approach to find C_2, C_3, C_4
 which will obtain values for the same concentrations.

$1400 C_2 - 100 C_4 = 909.1$
 $1100 C_2 - 1230 C_4 + 100 C_4 = 909.1$
 $1100 C_2 - 1130 C_4 = 909.1$
 $1100 C_2 - 1130 C_4 = 909.1$
 $1100 C_2 - 1130 C_4 = 909.1$

Confirmation Statement

$11400 > |1100| + |100|$ (yes)
 $11230 > |1100| + |100|$ (yes)
 $11320 > |100| + |1100|$ (yes)

is diagonally dominant.

order to solve the Gauss-Seidel equation, it must be modified.

Modification

$C_1 = \frac{1}{1.1} (909.1 + 100 C_2 - 0 C_4)$
 $C_2 = \frac{1}{1400} (-1100 C_2 - 100 C_4)$
 $C_3 = \frac{1}{1230} (-10 C_2 - 1100 C_4)$

$C_1 = \frac{1}{1.1} (909.1 + 100 C_2)$
 $C_2 = \frac{1}{1400} (-1100 C_2 - 100 C_4)$
 $C_3 = \frac{1}{1230} (-10 C_2 - 1100 C_4)$

$C_1 = \frac{1}{1.1} (909.1 + 100 C_2)$
 $C_2 = \frac{1}{1400} (-1100 C_2 - 100 C_4)$
 $C_3 = \frac{1}{1230} (-10 C_2 - 1100 C_4)$

Initial conditions

$C_1^0 = C_2^0 = C_3^0 = 1$

$1100 C_{A1} - 1100 C_{A2} = 1440 C_{A1} C_{A2} = 0$
 $1100 C_{A1} - 1100 C_{A2} - 1440 C_{A1} C_{A2} = 0$
 $1100 C_{A1} - 1300 C_{A2} = 0$ (4)

matrix of equations
 $C_{A0} - C_{A1} = 0.1 C_{A1}$ (1)
 $1000 C_{A1} + 100 C_{A2} = 1400 C_{A2}$ (2)
 $1000 C_{A1} + 100 C_{A2} = 1230 C_{A2}$ (3)
 $1100 C_{A1} = 1300 C_{A2}$ (4)

from eq (1)
 $C_{A0} - C_{A1} = 0.1 C_{A1}$
 $C_{A0} = 1.1 C_{A1}$
 $\therefore C_{A1} = \frac{1}{1.1} C_{A0}$

assume that C_{A0} is known to be 1 mol/litre
 $\therefore C_{A1} = 0.9091 \text{ mol/litre}$

which reduces the equation to three:
 $\therefore$ from eq (2)
 $1000 C_{A1} + 100 C_{A2} = 1400 C_{A2}$
 $1000 (0.9091) + 100 C_{A2} = 1400 C_{A2}$
 $909.1 + 100 C_{A2} = 1400 C_{A2}$

rearranging the equation
 $1000 C_{A1} - 100 C_{A2} = 909.1$ (1)
 $1100 C_{A1} - 1230 C_{A2} + 100 C_{A2} = 0$ (2)
 $1100 C_{A1} - 1130 C_{A2} = 0$ (3)

using Gauss seidel approach to find C_{A2} , C_{A1} , C_{A3}
 which will obtain values for the steady state concentrations

matrix of equations
 $1000 C_{A1} - 100 C_{A2} = 909.1$
 $1100 C_{A1} - 1230 C_{A2} + 100 C_{A2} = 0$
 $1100 C_{A1} - 1130 C_{A2} = 0$

dominant diagonal
 $1000 > |100| + |0|$ (yes)
 $1100 > |1230| + |100|$ (yes)
 $1130 > |100| + |1100|$ (yes)

iteration statement
 iter diagonally dominant

order to solve the Gauss-seidel equation, it must be modified.

matrix of equations
 $C_{A1}(n) = \frac{1}{1400} (909.1 + 100 C_{A2}(n-1) - 0 C_{A3}(n-1))$
 $C_{A2}(n) = \frac{1}{1230} (-1100 C_{A1}(n) - 100 C_{A3}(n-1))$
 $C_{A3}(n) = \frac{1}{-1300} (-100 C_{A2}(n) - 1100 C_{A1}(n))$

$C_{A1}(n) = \frac{1}{1400} (909.1 + 100 C_{A2}(n-1))$
 $C_{A2}(n) = \frac{1}{-1230} (-1100 C_{A1}(n) - 100 C_{A3}(n-1))$
 $C_{A3}(n) = \frac{1}{-1300} (-100 C_{A2}(n) - 1100 C_{A1}(n))$

assume initial conditions
 $C_{A1}(0) = C_{A2}(0) = C_{A3}(0) = 1$

1st iteration with $r=1$

$$C_2(1) = \frac{1}{1400} (909.1 + 1000 C_3(1))$$

$$C_3(1) = \frac{1}{1400} (909.1 + 100 C_1(1))$$

$$C_2(1) = 0.7208$$

$$C_3(1) = \frac{1}{-1230} (-1100 C_2(1) - 100 C_4(1))$$

$$C_3(1) = \frac{1}{-1230} (-1100 (0.7208) - 100 C_4(1))$$

$$C_3(1) = 0.7259$$

$$C_4(1) = \frac{1}{-1300} (-1100 C_3(1))$$

$$C_4(1) = \frac{1}{-1300} (-1100 (0.7259))$$

$$C_4(1) = 0.6142$$

2nd iteration with $r=2$

$$C_2(2) = 0.7208, C_3(2) = 0.7259, C_4(2) = 0.6142$$

$$C_2(2) = \frac{1}{1400} (909.1 + 1000 C_3(2))$$

$$C_3(2) = \frac{1}{1400} (909.1 + 100 (0.7259))$$

$$C_2(2) = 0.7202$$

$$C_3(2) = \frac{1}{-1230} (-1100 C_2(2) - 100 C_4(2))$$

$$C_3(2) = \frac{1}{-1230} (-1100 (0.7202) - 100 (0.6142))$$

$$C_3(2) = 0.6720$$

$$C_4(2) = \frac{1}{-1300} (-1100 C_3(2))$$

$$C_4(2) = \frac{1}{-1300} (-1100 (0.6720))$$

$$C_4(2) = 0.5686$$

3rd iteration with $r=3$

$$C_2(3) = 0.7202, C_3(3) = 0.6720, C_4(3) = 0.5686$$

$$C_2(3) = \frac{1}{1400} (909.1 + 1000 C_3(3))$$

$$C_2(3) = \frac{1}{1400} (909.1 + 100 (0.6720))$$

$$C_2(3) = 0.6994$$

$$C_3(3) = \frac{1}{-1230} (-1100 C_2(3) - 100 C_4(3))$$

$$C_3(3) = \frac{1}{-1230} (-1100 (0.6994) - 100 (0.5728))$$

$$C_3(3) = 0.6720$$

$$C_4(3) = \frac{1}{-1300} (-1100 C_3(3))$$

$$C_4(3) = \frac{1}{-1300} (-1100 (0.6720))$$

$$C_4(3) = 0.5686$$

4th iteration with $r=4$

$$C_2(4) = 0.6994, C_3(4) = 0.6720, C_4(4) = 0.5686$$

$$C_2(4) = \frac{1}{1400} (909.1 + 1000 C_3(4))$$

$$C_2(4) = \frac{1}{1400} (909.1 + 100 (0.6720))$$

$$C_2(4) = 0.6993$$

$$c_3^{(5)} = \frac{1}{-1230} (-1100c_2^{(4)} - 100c_4^{(4)})$$

$$c_3^{(4)} = \frac{1}{-1230} (-1100(0.6973) - 100(0.5686))$$

$$c_3^{(4)} = 0.6698$$

$$c_4^{(4)} = \frac{1}{-1300} (-1100c_3^{(4)})$$

$$c_4^{(4)} = \frac{1}{-1300} (-1100(0.6698))$$

$$c_4^{(4)} = 0.5668$$

5th iteration when $r=5$

$$c_2^{(5)} = 0.6973, c_3^{(4)} = 0.6698, c_4^{(4)} = 0.5668$$

$$c_2^{(5)} = \frac{1}{1400} (909.1 + 100c_3^{(4)})$$

$$c_2^{(5)} = \frac{1}{1400} (909.1 + 100(0.6698))$$

$$c_2^{(5)} = 0.6972$$

$$\rightarrow c_3^{(5)} = \frac{1}{-1230} (-1100c_2^{(5)} - 100c_4^{(4)})$$

$$c_3^{(5)} = \frac{1}{-1230} (-1100(0.6972) - 100(0.5668))$$

$$c_3^{(5)} = 0.6696$$

$$\Rightarrow c_4^{(5)} = \frac{1}{-1300} (-1100c_3^{(5)})$$

$$c_4^{(5)} = \frac{1}{-1300} (-1100(0.6696))$$

$$c_4^{(5)} = 0.5666$$

6th iteration when $r=6$

$$c_2^{(6)} = \frac{1}{1400} (909.1 + 100c_3^{(5)})$$

$$c_2^{(6)} = \frac{1}{1400} (909.1 + 100(0.6696))$$

$$c_2^{(6)} = 0.6972$$

$$c_3^{(6)} = \frac{1}{-1230} (-1100c_2^{(6)} - 100c_4^{(5)})$$

$$c_3^{(6)} = \frac{1}{-1230} (-1100(0.6972) - 100(0.5666))$$

$$c_3^{(6)} = 0.6696$$

$$c_4^{(6)} = \frac{1}{-1300} (-1100c_3^{(6)})$$

$$c_4^{(6)} = \frac{1}{-1300} (-1100(0.6696))$$

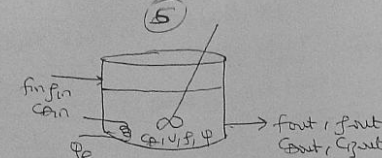
$$c_4^{(6)} = 0.5666$$

In further iterations (e.g., 7th, 8th, 9th, 10th) will attain constant value for c_2, c_3 , and c_4 . Hence iteration terminated here.

	0	1	2	3	4	5	6
c_2		0.7238	0.712	0.6974	0.6973	0.6972	0.6972
c_3		0.7239	0.6770	0.6720	0.6698	0.6696	0.6696
c_4		0.6142	0.5725	0.5686	0.5668	0.5666	0.5666

Recall, $c_2 = 0.6972$

$$c = \begin{pmatrix} 0.6972 \\ 0.6696 \\ 0.5666 \end{pmatrix}$$
 limiting values to 3



Process
A stirred tank reactor

System description
The system is a stirred tank reactor. The liquid in the tank is perfectly mixed, and the liquid is assumed to be perfectly mixed. Heat is transferred from the system to the surroundings and the heat generated within the system is removed.

Assumptions

- 1) Since the liquid is considered perfectly mixed, hence concentration is assumed throughout the reactor. $\therefore F_{in} = F_{out} = F$
- 2) A constant volume reactor i.e. $F_{in} = F_{out}$
- 3) Assumed CSTR or perfectly mixed, $C_{out} = C$, $C_F = C_{in}$ and $T = T_{out}$

Mathematical Formulation
(Non-isothermal conditions)

Mass Balance

$$\left[\text{Rate of mass in} \right] - \left[\text{Rate of mass out} \right] = \left[\text{Rate of mass accumulation} \right]$$

$$F_{in} C_{in} - F_{out} C_{out} = \frac{d}{dt} (FV)$$

Since $F_{in} = F_{out} = F$

$$F(C_{in} - C_{out}) = \frac{dV}{dt}$$

$$\therefore F_{in} - F_{out} = \frac{dV}{dt} \quad \text{--- (1)}$$

$F_{in} - F_{out} = 0$
 $\therefore F_{in} = F_{out}$

Component Balance

$$\left[\text{Rate of mass of A in} \right] - \left[\text{Rate of mass of A out} \right] - \left[\text{Rate of consumption of A due to reaction} \right] = \left[\text{Rate of accumulation of A} \right]$$

$$F_{in} C_{in} - F_{out} C_{out} - (-r_A)V = \frac{d}{dt} (C_A V) \quad \text{--- (2)}$$

Since $F_{in} = F_{out}$

$$F_{in} C_{in} - F_{in} C_{out} - (-r_A)V = V \frac{dC_A}{dt}$$

Also $C_A = C_{out}$ and $-r_A = k C_A$ (for first-order reaction)

$$F_{in} C_{in} - F_{in} C_A - k C_A V = V \frac{dC_A}{dt}$$

$$V \frac{dC_A}{dt} = F_{in} C_{in} - F_{in} C_A - k C_A V \quad \text{--- (4)}$$

Steady State By V

$$\frac{dC_A}{dt} = \frac{F_{in}}{V} C_{in} - \frac{F_{in}}{V} C_A - k C_A$$

$$\frac{dC_A}{dt} + \frac{F_{in}}{V} C_A + k C_A = \frac{F_{in}}{V} C_{in}$$

$$\frac{dC_A}{dt} + \left(\frac{F_{in}}{V} + k \right) C_A = \frac{F_{in}}{V} C_{in} \quad \text{--- (5)}$$

Eq (5), component balance and dynamic behaviour of the reactor for non-isothermal conditions. Under isothermal conditions, there is variation in temperature. Hence, the requirement of energy balance

$$\left[\begin{matrix} \text{Rate of} \\ \text{energy} \\ \text{in} \end{matrix} \right] - \left[\begin{matrix} \text{Rate of} \\ \text{energy} \\ \text{out} \end{matrix} \right] + \left[\begin{matrix} \text{Rate of} \\ \text{energy} \\ \text{generated} \end{matrix} \right] - \left[\begin{matrix} \text{Rate of} \\ \text{energy} \\ \text{exchanged} \end{matrix} \right] = \left[\begin{matrix} \text{Rate of} \\ \text{energy} \\ \text{accumulated} \end{matrix} \right]$$

$$\pm (p_2 - u_2) = \frac{d}{dt} \psi(\hat{u} + \hat{i} + \hat{p}) \quad \text{--- (6)}$$

Assumptions

- 1) Heat of reaction is exothermic $\Rightarrow -Q_R$
- 2) Heat generated $Q_R = (-\Delta H_r (-m)) \sqrt{}$
- 3) Kinetic and potential energies negligible
- 4) No work done $\therefore W_b = 0$, and $Q = \dot{h}$
- 5) $\dot{h} = \dot{Q} (T - T_{ref})$

$$\therefore \hat{h}_{in} = \hat{C}_p (T_{in} - T_{ref})$$

$$\hat{h}_{out} = \hat{c}_p (T_{out} - T_{ref}) \text{ or } \hat{c}_p (T - T_{ref})$$

case Test 27

6) $f_{in} = f_{out} = f$.

$$\text{Anfangshin} - \text{Endwert} + (-\Delta H - (-P)V) = Q_e$$

$$= \frac{d}{dt} \text{fwh} \quad \text{--- (7)}$$

where $\eta = 1/\cos\theta$

$$\frac{d}{dt} \left(\rho \hat{C}_p (T - T_{ref}) \right) = \dot{Q} - \dot{W} + \dot{m} \hat{C}_p (T_{in} - T_{ref}) - \dot{m} \hat{C}_p (T_{out} - T_{ref}) + \dot{m} \hat{C}_p (T_{ref} - T_{ref})$$

Dividing through by $f \hat{c}^p$ since $f_{in} = f_{out} = f$

$$\frac{\text{Kinf} \cdot P \cdot (T_{in} - T_{ref})}{\rho \cdot C_p} - \frac{\text{Kout} \cdot P_{out} \cdot C_p \cdot (T_{ref} - T_{in})}{\rho \cdot C_p} + \frac{(T_{in} - T_{ref})}{\rho \cdot C_p}$$

$$= \frac{d}{dt} \left(\frac{1}{f} \frac{dV}{dt} (T - T_{ref}) \right) + \frac{(-DB + KC_{DPO})}{f} - \frac{f}{f}$$

$$= \sqrt{\frac{dT}{dt}} + T \frac{d}{dt} \sqrt{\frac{dT}{dt}} = \sqrt{\frac{dT}{dt}} + T \frac{1}{2} \frac{1}{\sqrt{\frac{dT}{dt}}} \frac{dT}{dt} = \sqrt{\frac{dT}{dt}} + \frac{T}{2} \frac{dT}{dt} \sqrt{\frac{dT}{dt}}$$

Because volume is constant
- T_{ref} - P_{ref} ($T =$)

$$\therefore R_{in} = R_{out} = R$$

$$\frac{dT}{dt} = P_{in} T_{in} - P_{out} T_{out} - \frac{A U (T_{out} - T_{amb})}{R_{ext}} - \frac{P_{el}}{R_{ext}}$$

$$\frac{d}{dt} = \frac{p}{V} \frac{dV}{dt} - \frac{D \ln K_{eq}}{R \ln K_{eq}} - \frac{D}{V P \hat{C}_P} \quad (9)$$

$$+ \left(\frac{P}{V} \right)_T = \left(\frac{P}{V} \right)_{T_0} - \frac{\Delta H_{\text{vap}}}{R T_0^2} - \frac{P_0}{V P_0^2}$$

FOR SOLUBILITY (COEFFICIENT OF PRESSION ANALYSIS) $\rightarrow (P)$

parameter of constant value $2K_1 V$

Known variables = F_{in}, C_{in}
 remaining variables = $C_A = H_V$ (number of variables)
 $\therefore$ No of equations $H_E = 1$
 $\therefore H_E = H_V - H_K$
 $H_E = 1 - 1 = 0$
 Hence the model equation at non-isothermal condition
 can be solved given the initial conditions of
 $C_A(t_i) = C_{A_i}$

DEGREE OF FREEDOM ANALYSIS FOR EX 10
 Parameters of constant value = C_p, ρ, A_{HT}, V
 Fixed variables = T_{in}, P_{in}, C_{Ain}
 remaining variables = $T, C_A, C_B = 3$
 No of equations $H_E = 2$
 $\therefore H_E = H_V - H_K$
 $= 4 - 2 = 2$
 Since the degree of freedom is 2, two extra
 equations are

(i) $Q_R = U A_{HT} (T - T_a)$
 (ii) $K = A_{HT} P \left(\frac{Q}{RT} \right)$
 where T_a = temperature of air
 A_{HT} = overall heat transfer area.

(6)

Assumptions
 At steady state $\therefore$ Accumulation, $\frac{dC_A}{dt} = 0$
 $V \frac{dC_A}{dt} = F C_{Ain} - F C_A - (-r_A) V$

$\frac{dC_A}{dt} = \frac{F}{V} C_{Ain} - \frac{F}{V} C_A - (-r_A)$
 Given that reaction is of 3rd order $\therefore -r_A = k C_A^3$
 $0 = \frac{F}{V} C_{Ain} - \frac{F}{V} C_A - k C_A^3$
 with initial conditions, $P = 10 \text{ m}^2/\text{s}, k = 2 \text{ mol}^{-2} \text{ m}^2 \text{ s}^{-1}$
 $V = 25 \text{ m}^3, C_{Ain} = 0.1 \text{ mol/m}^3$
 $0 = \frac{(10)(0.1)}{25} - \frac{(10)}{25} C_A - (2) C_A^3$
 $0 = 0.04 - 0.4 C_A - 2 C_A^3$

EXPANDING
 $C_A^3 + 0.4 C_A - 0.04 = 0$
 $f(C_A) = 2 C_A^3 + 0.4 C_A - 0.04$
 and $f'(C_A) = 6 C_A^2 + 0.4$

Applying Newton-Raphson's method
 at an initial value of $C_A = 1$
 $C_{A1} = C_{A1} - \frac{f(C_{A1})}{f'(C_{A1})}$

$C_{A1} = 2(1)^3 + 0.4(1) - 0.04 = 2.36$
 $C_{A1} = 6.4$
 $C_{A2} = 1 - \frac{2.36}{6.4}$
 $C_{A2} = 1 - 0.36875$
 $C_{A2} = 0.6313 \text{ mol/m}^3$

when $C_{A0} = 2$

$$C_A = C_{A0} - \frac{f(2)}{f'(2)}$$

$$f(2) = 2(2)^3 + 0.4(2) - 0.04 = 16.76$$

$$f'(2) = 6(2)^2 + 0.4 = 24.4$$

$$C_A = 2 - \frac{16.76}{24.4}$$

$$C_A = 2 - 0.6869$$

$$C_A = 1.3131 \text{ mol/m}^3$$

when $C_{A0} = 3$

$$C_A = C_{A0} - \frac{f(3)}{f'(3)}$$

$$f(3) = 2(3)^3 + 0.4(3) - 0.04 = 55.16$$

$$f'(3) = 6(3)^2 + 0.4 = 54.4$$

$$C_A = 3 - \frac{55.16}{54.4}$$

$$C_A = 3 - 1.01397$$

$$C_A = 1.9860 \text{ mol/m}^3$$

when $C_{A0} = 4$

$$C_A = C_{A0} - \frac{f(4)}{f'(4)}$$

$$f(4) = 2(4)^3 + 0.4(4) - 0.04 = 129.56$$

$$f'(4) = 6(4)^2 + 0.4 = 96.4$$

$$C_A = 4 - \frac{129.56}{96.4}$$

$$C_A = 4 - 1.3439$$

$$C_A = 2.6560 \text{ mol/m}^3$$

$$\text{When } C_{A0} = 5$$

$$C_A = C_{A0} - \frac{f(S)}{f'(S)}$$

$$f(S) = 2(S)^3 + 0.4(S) - 0.04 = 251.96$$

$$f'(S) = 6(S)^2 + 0.4 = 150.4$$

$$C_A = 5 - \frac{251.96}{150.4}$$

$$C_A = 5 - 1.6753$$

$$C_A = 3.3247 \text{ mol/m}^3$$

$0.028 \frac{LA}{V} = 2.5688 \times 10^{-3} - 0.000784$
 $\frac{LA}{V} = 2.7848 \times 10^{-3}$
 $\frac{LA}{V} = \frac{2.7848 \times 10^{-3}}{0.028}$
 $\boxed{\frac{LA}{V} = 0.09950}$
 $\therefore \frac{L_{MA}}{V} = 0.09950, \frac{A^2}{r_{HV}} = 4.635 \times 10^{-6}$

How long it would take to reach a pressure of 1814 Pa

$\frac{dV}{dt} = 0.0002 \text{ m}^3/\text{s}$
 $\therefore V = 0.0002t$
 Substituting into the model equation
 $V^2 + \frac{LA}{V} V = \frac{(-\Delta p) A^2 t}{r_{HV}}$
 $(0.0002t)^2 + \frac{LA}{V} (0.0002t) = \frac{(-\Delta p) A^2 t}{r_{HV}}$
 $4.0 \times 10^{-8} t^2 + 0.0002 \frac{LA}{V} t = \frac{A^2}{r_{HV}} (-\Delta p) t$
 When $(-\Delta p) = 1814 \text{ Pa}$
 $4.0 \times 10^{-8} t^2 + 0.0002 (0.09950) t = 4.635 \times 10^{-6} + (18) t$
 $4.0 \times 10^{-8} t^2 + 1.99 \times 10^{-5} t = 8.343 \times 10^{-5} t$
 $4.0 \times 10^{-8} t^2 + 1.99 \times 10^{-5} t - 8.343 \times 10^{-5} t = 0$
 $4.0 \times 10^{-8} t^2 - 6.353 \times 10^{-5} t = 0$
 $4.0 \times 10^{-8} t = 6.353 \times 10^{-5}$

$$4.0 \times 10^{-8} = 0.853 \times 10^{-4} \\ t = \frac{6.353 \times 10^{-5}}{4.0 \times 10^{-8}}$$

$$t = 1588.35 \text{ (26.5 mins)}$$

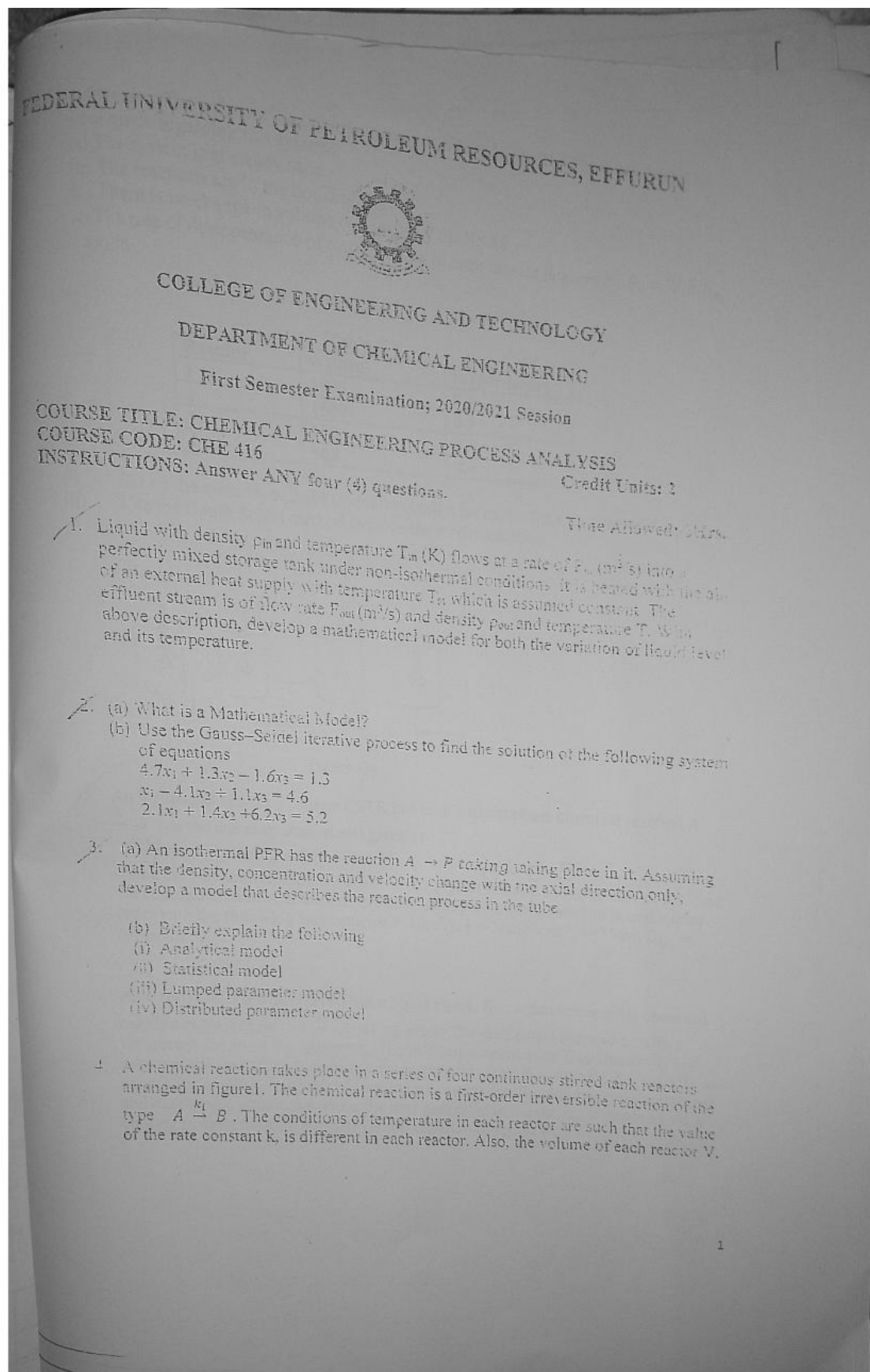
∴ the volume of the filtrate collected at this stage
at $t = 1588.35$

from the expression $V = 0.0002t$

$$V = 0.0002 \times 1588.35$$

$$V = 0.3177 \text{ m}^3$$

∴ the volume of the filtrate collected at the end of
 $1588.35 \text{ (26.5 mins)} = 0.3177 \text{ m}^3$



is different. The values of k_i and v_i are given in Table 1. The following assumptions were made regarding this system:

- The system is at steady state.
- The reactions are in the liquid phase.
- There is no change in volume or density of the liquid.
- The rate of disappearance of component A in each reactor is given by $R_i = V_i k_i C_{A_i}$ mol/h

Table Q4

Reactor	v_i (litre)	k_i (per hour)
1	1000	0.1
2	1500	0.2
3	100	0.3
4	500	0.4

- Set up the material balance (model) equation for each of the four reactors.
- Use the Gauss Seidel method to solve the model equation

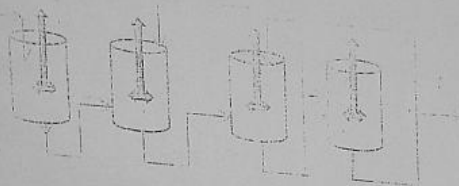


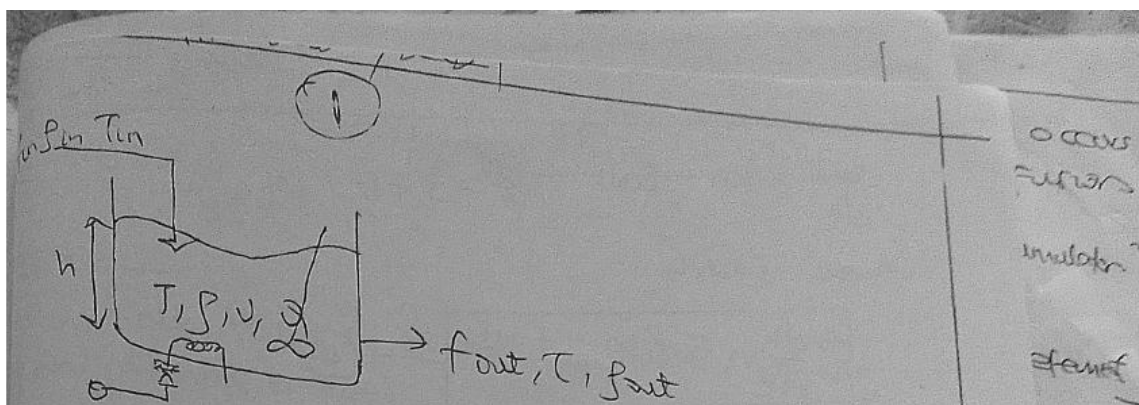
Figure Q4

- An isothermal constant volume CSTR in which a liquid phase chemical reaction $A \xrightarrow{k} B$ has the unsteady state model given as

$$V \frac{dC_A}{dt} = f C_{A_{in}} - f C_A - (-r_A) V.$$

Using the Newton Raphson's method for a range of 0.1 to 0.5 and an initial root of 0.1 , find the concentration of A at the exit stream at steady state if the reaction is in the third order with the following assumptions: $f=10 \text{ m}^3/\text{s}$, $k=2 \text{ mol}^{-2} \text{ m}^2 \text{ s}^{-1}$, $v=25 \text{ m}^3$, $C_{A_{in}} = 0.1 \text{ mol/m}^3$.

- A perfectly mixed tank (CSTR) has a liquid phase, first order, irreversible chemical reaction expressed as $A \xrightarrow{k_1} B$ taking place. The feed enters the reactor with a volumetric flow rate F_{in} , density, ρ_{in} (kg/m^3) and concentration $C_{A_{in}}$ (mol/m^3). The output comes out of the reactor at a volumetric flow rate F_{out} , density, ρ_{out} (kg/m^3) and concentration, $C_{A_{out}}$ (mol/m^3). Assuming a constant density, volume, non-isothermal and exothermic conditions where the heat generated in the reactor is removed via a cooling system, develop a model for the variation of specie A.



SYSTEM CONFIGURATION

The system is the liquid in the tank having non-constant condition and it is perfectly mixed, the liquid enters the tank and takes liquid out of the tank as well as interacting with the surrounding. The liquid is being heated with a constant temperature from an external heat supply.

ASSUMPTIONS

The system is perfectly mixed so the density of the liquid stream is the same as that of its influent. The tank system is non-isothermal.

MODEL FORMULATION

Mass Balance

$$\left[\text{Rate of mass in} \right] - \left[\text{Rate of mass out} \right] = \left[\text{Rate of mass accumulation} \right]$$

$$\dot{m}_{in} - \dot{m}_{out} = \frac{d}{dt} \rho V \quad (1)$$

$$\dot{m}_{in} = \dot{m}_{out} = \dot{m} \quad (\text{assumed})$$

$$\rho \frac{dV}{dt} = \dot{m}_{in} - \dot{m}_{out} \quad (2)$$

$$\rho \frac{dV}{dt} = \dot{m} (f_{in} - f_{out})$$

$$\frac{du}{dt} = f_{in} - f_{out} \quad (3)$$

where $V = Ah$

$$\therefore \boxed{A \frac{dh}{dt} = f_{in} - f_{out}} \quad (4)$$

equation (4) is a mass balance equation for the system,
DEGREE OF FREEDOM ANALYSIS

parameters of constant value = A, g
 variables whose values can be externally fixed = f_{in}
 Remaining unknown variables = $h, f_{out} \therefore N_U = 2$
 Number of equations (N_E) = 1

$$\therefore \text{degree of freedom } N_D = N_U - N_E$$

$$= 2 - 1 = 1$$

Hence for the system to be exactly specified, we need
 therefore one more equation, this extra equation
 relates out flow rate f_{out} to the liquid level h

$$\boxed{f_{out} = \alpha \sqrt{h}}$$

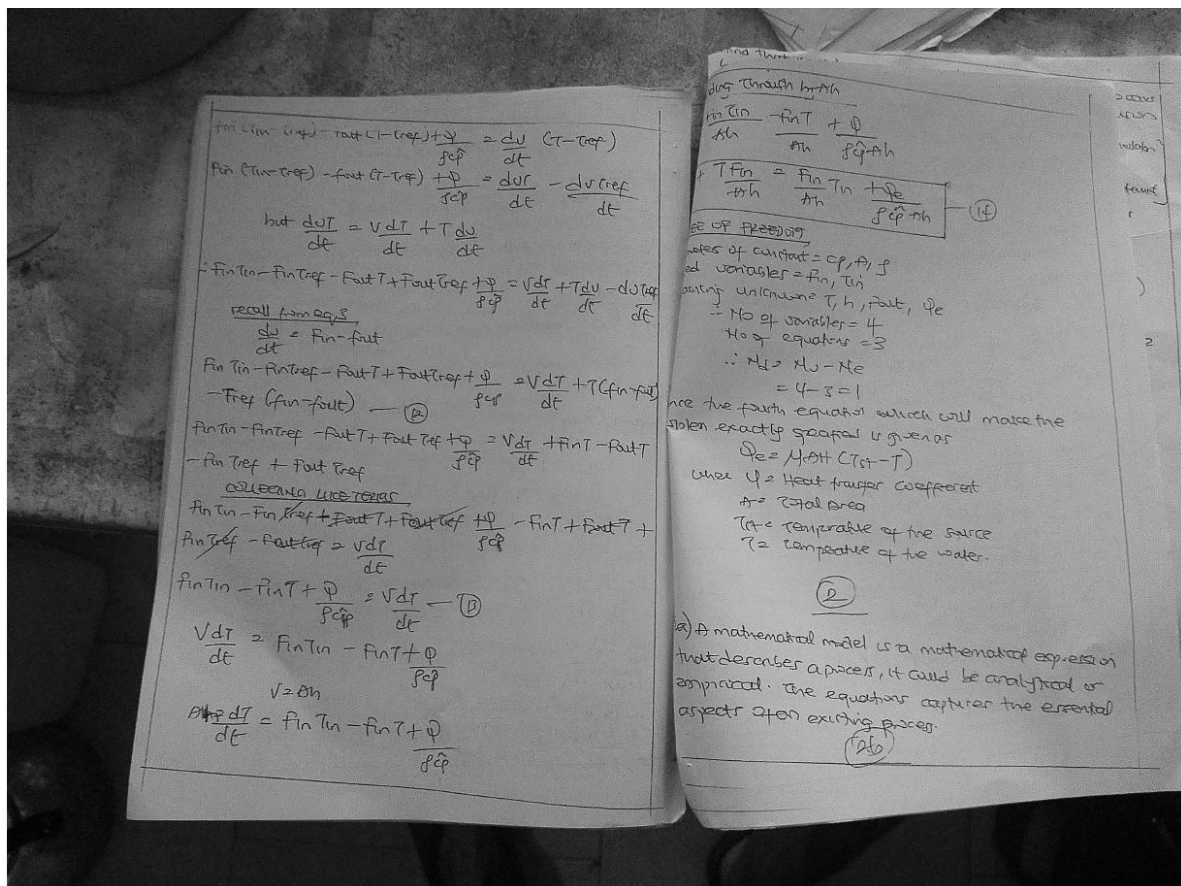
where α = discharge coefficient

$\therefore$ the model equation for the variation of liquid
 level is

$$\boxed{A \frac{dh}{dt} + f_{out} = f_{in}} \quad \text{where } f_{out} = \alpha \sqrt{h} \quad (5)$$

ENERGY BALANCE

$$\left[\begin{array}{c} \text{rate of} \\ \text{energy} \\ \text{in} \end{array} \right] - \left[\begin{array}{c} \text{rate of} \\ \text{energy} \\ \text{out} \end{array} \right] + \left[\begin{array}{c} \text{rate of} \\ \text{energy} \\ \text{generated} \end{array} \right] - \left[\begin{array}{c} \text{rate of} \\ \text{exchange} \\ \text{of energy} \\ \text{with} \\ \text{surroundings} \end{array} \right] = \left[\begin{array}{c} \text{rate of} \\ \text{energy} \\ \text{accumulated} \end{array} \right]$$



$$\dot{m}_w (T_w - T_{ref}) - \dot{m}_a (T_a - T_{ref}) + \frac{Q}{\rho C_p} = \frac{dU}{dt} (T - T_{ref})$$

$$\dot{m}_w (T_w - T_{ref}) - \dot{m}_a (T_a - T_{ref}) + \frac{Q}{\rho C_p} = \frac{dU}{dt} - \frac{dU_{ref}}{dt}$$

$$\text{but } \frac{dU}{dt} = \frac{V dT}{dt} + T \frac{dV}{dt}$$

$$\dot{m}_w T_w - \dot{m}_a T_a - \dot{m}_a T_{ref} + \dot{m}_a T_{ref} + \frac{Q}{\rho C_p} = \frac{V dT}{dt} + T \frac{dV}{dt} - \dot{m}_a T_{ref}$$

recall Air eqs

$$\frac{dV}{dt} = \dot{m}_w - \dot{m}_a$$

$$\dot{m}_w T_w - \dot{m}_a T_a - \dot{m}_a T_{ref} + \dot{m}_a T_{ref} + \frac{Q}{\rho C_p} = \frac{V dT}{dt} + T (\dot{m}_w - \dot{m}_a) - \dot{m}_a T_{ref}$$

$$\dot{m}_w T_w - \dot{m}_a T_a - \dot{m}_a T_{ref} + \dot{m}_a T_{ref} + \frac{Q}{\rho C_p} = \frac{V dT}{dt} + \dot{m}_w T - \dot{m}_a T$$

$$-\dot{m}_a T_{ref} + \dot{m}_a T_{ref}$$

collecting like terms

$$\dot{m}_w T_w - \dot{m}_a T_a + \dot{m}_a T_{ref} + \frac{Q}{\rho C_p} - \dot{m}_w T + \dot{m}_a T = \frac{V dT}{dt}$$

$$\dot{m}_w T_w - \dot{m}_a T_a + \frac{Q}{\rho C_p} = \frac{V dT}{dt}$$

$$\frac{V dT}{dt} = \dot{m}_w T_w - \dot{m}_a T_a + \frac{Q}{\rho C_p}$$

$$\frac{dV}{dt} = \dot{m}_w - \dot{m}_a$$

$$\frac{dU}{dt} = \dot{m}_w T_w - \dot{m}_a T_a + \frac{Q}{\rho C_p}$$

$(4.7)(1.3x_2 - 1.6x_3) = 1.3$
 $x_1 - 4.1x_2 + 1.1x_3 = 4.6$
 $2.1x_1 + 1.4x_2 + 6.2x_3 = 5.2$

Testing for Diagonal Dominance

$1.47 > 1.3 + 1.16$ (yes)
 $1.41 > 1.1 + 1.1$ (yes)
 $1.62 > 1.2 + 1.1$ (yes)

It is diagonally dominant

Hence using

$$x_1^{(r)} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^{(r-1)} - a_{13}x_3^{(r-1)})$$

$$x_2^{(r)} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^{(r)} - a_{23}x_3^{(r-1)})$$

$$x_3^{(r)} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^{(r)} - a_{32}x_2^{(r)})$$

Initial conditions rule of $x_1^{(0)} = x_2^{(0)} = x_3^{(0)} = 1$

∴ when $r=1$

$$x_1^{(1)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(0)} + 1.6x_3^{(0)})$$

$$x_1^{(1)} = \frac{1}{4.7} (1.3 - 1.3(1) + 1.6(1))$$

$$x_1^{(1)} = 0.3404$$

$$x_2^{(1)} = \frac{1}{-4.1} (4.6 - 1x_1^{(1)} - 1.1x_3^{(0)})$$

$$x_2^{(1)} = \frac{1}{-4.1} (4.6 - 1(0.3404) - 1.1(1))$$

$$x_2^{(1)} = -0.7706$$

when $r=2$ [2nd iteration]

$$x_1^{(2)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(1)} + 1.6x_3^{(1)}) = 0.8974$$

$$x_2^{(2)} = \frac{1}{-4.1} (4.6 - 1x_1^{(2)} - 1.1x_3^{(1)})$$

$$x_2^{(2)} = \frac{1}{-4.1} (4.6 - 1(0.8974) - 1.1(0.8974))$$

$$x_2^{(2)} = -0.6876$$

$$x_3^{(2)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(2)} - 1.4x_2^{(2)})$$

$$x_3^{(2)} = \frac{1}{6.2} (5.2 - 2.1(0.8974) - 1.4(-0.6876))$$

$$x_3^{(2)} = 0.7251$$

3rd iteration when $r=3$

$$x_1^{(3)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(2)} + 1.6x_3^{(2)})$$

$$x_1^{(3)} = \frac{1}{4.7} (1.3 - 1.3(-0.6876) + 1.6(0.7251))$$

$$x_1^{(3)} = 0.7132$$

$$x_1^{(3)} = \frac{1}{-4.1} [5.6 - 1.1x_2^{(2)} - 1.1x_3^{(2)}]$$

$$x_2^{(3)} = \frac{1}{-4.1} [4.6 - 1(0.7136) - 1.1(0.7251)]$$

$$x_2^{(3)} = \frac{1}{-4.1} (3.08879)$$

$$x_2^{(3)} = -0.7534$$

$$x_3^{(3)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(3)} - 1.4x_2^{(2)})$$

$$x_3^{(3)} = \frac{1}{6.2} (5.2 - 2.1(0.7136) - 1.4(-0.7534))$$

$$x_3^{(3)} = 0.7671$$

4th iteration when $r=4$

$$x_1^{(3)} = 0.7136, x_2^{(3)} = -0.7534, x_3^{(3)} = 0.7671$$

$$x_1^{(4)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(3)} + 1.6x_3^{(3)})$$

$$x_1^{(4)} = \frac{1}{4.7} (1.3 - 1.3(-0.7534) + 1.6(0.7671))$$

$$x_1^{(4)} = 0.7461$$

$$x_2^{(4)} = \frac{1}{-4.1} [4.6 - 1x_1^{(4)} - 1.1x_3^{(3)}]$$

$$x_2^{(4)} = \frac{1}{-4.1} [4.6 - 1(0.7461) - 1.1(0.7671)]$$

$$x_2^{(4)} = -0.7342$$

$$x_3^{(4)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(4)} - 1.4x_2^{(4)})$$

$$x_3^{(4)} = \frac{1}{6.2} (5.2 - 2.1(0.7461) - 1.4(-0.7342))$$

$$x_3^{(4)} = 0.7518$$

and that mass transfer in the differential
by two mechanisms which are -
Rate of -

5th iteration when $r=5$

$$x_1^{(5)} = 0.7461, x_2^{(5)} = -0.7342, x_3^{(5)} = 0.7518$$

$$x_1^{(5)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(5)} + 1.6x_3^{(5)})$$

$$x_1^{(5)} = \frac{1}{4.7} (1.3 - 1.3(-0.7342) + 1.6(0.7518))$$

$$x_1^{(5)} = 0.7856$$

$$x_2^{(5)} = \frac{1}{-4.1} (4.6 - 1.1x_1^{(5)} - 1.1x_3^{(5)})$$

$$x_2^{(5)} = \frac{1}{-4.1} (4.6 - 1(0.7856) - 1.1(0.7518))$$

$$x_2^{(5)} = -0.7408$$

$$x_3^{(5)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(5)} - 1.4x_2^{(5)})$$

$$x_3^{(5)} = \frac{1}{6.2} (5.2 - 2.1(0.7856) - 1.4(-0.7408))$$

$$x_3^{(5)} = 0.7568$$

6th iteration when $r=6$

$$x_1^{(6)} = 0.7856, x_2^{(6)} = -0.7408, x_3^{(6)} = 0.7568$$

$$x_1^{(6)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(6)} + 1.6x_3^{(6)})$$

$$x_1^{(6)} = \frac{1}{4.7} (1.3 - 1.3(-0.7408) + 1.6(0.7568))$$

$$x_1^{(6)} = 0.7891$$

$$x_2^{(6)} = \frac{1}{-4.1} (4.6 - 1.1x_1^{(6)} - 1.1x_3^{(6)})$$

$$x_2^{(6)} = \frac{1}{-4.1} (4.6 - 1(0.7891) - 1.1(0.7568))$$

$$x_2^{(6)} = -0.7386$$

$$\Rightarrow x_3^{(6)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(6)} - 1.4x_2^{(6)})$$

$$x_3^{(6)} = \frac{1}{6.2} (5.2 - 2.1(0.7341) - 1.4(-0.7386))$$

$$x_3^{(6)} = 0.7552$$

7th iteration when $r=7$

$$x_1^{(6)} = 0.7391, x_2^{(6)} = -0.7386, x_3^{(6)} = 0.7552$$

$$\Rightarrow x_1^{(7)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(6)} + 1.6x_3^{(6)})$$

$$x_1^{(7)} = \frac{1}{4.7} (1.3 - 1.3(-0.7386) + 1.6(0.7552))$$

$$x_1^{(7)} = 0.7380$$

$$\Rightarrow x_2^{(7)} = \frac{1}{-4.1} (4.6 - 1x_1^{(7)} - 1.1x_3^{(6)})$$

$$x_2^{(7)} = \frac{1}{-4.1} (4.6 - 1(0.7380) - 1.1(0.7552))$$

$$x_2^{(7)} = -0.7393$$

$$\Rightarrow x_3^{(7)} = \frac{1}{6.2} (5.2 - 2.1x_1^{(7)} - 1.4x_2^{(7)})$$

$$x_3^{(7)} = \frac{1}{6.2} (5.2 - 2.1(0.7380) - 1.4(-0.7393))$$

$$x_3^{(7)} = 0.7557$$

8th iteration when $r=8$

$$x_1^{(7)} = 0.7380, x_2^{(7)} = -0.7393, x_3^{(7)} = 0.7557$$

$$\Rightarrow x_1^{(8)} = \frac{1}{4.7} (1.3 - 1.3x_2^{(7)} + 1.6x_3^{(7)})$$

$$x_1^{(8)} = \frac{1}{4.7} (1.3 - 1.3(-0.7393) + 1.6(0.7557))$$

$$x_1^{(8)} = 0.7383$$

$$\Rightarrow x_2^{(8)} = \frac{1}{-4.1} (4.6 - 1x_1^{(8)} - 1.1x_3^{(7)})$$

$$x_2^{(8)} = \frac{1}{-4.1} (4.6 - 1(0.7383) - 1.1(0.7557))$$

$$x_2^{(8)} = -0.7391$$

and that mass transfer is by two means...

$$x_1(8) = \frac{1}{6.2} (5.2 - 2.1x_1(8) - 1.4x_2(8))$$

$$x_1(8) = \frac{1}{6.2} (5.2 - 2.1(0.7383) - 1.4(-0.7391))$$

$$x_1(8) = 0.7555$$

9th iteration when $r=9$

$$x_1(8) = 0.7383, x_2(8) = -0.7391, x_3(8) = 0.7555$$

$$x_1(9) = \frac{1}{4.7} (1.3 - 1.3x_2(8) + 1.6x_3(8))$$

$$x_1(9) = \frac{1}{4.7} (1.3 - 1.3(-0.7391) + 1.6(0.7555))$$

$$x_1(9) = 0.7382$$

$$x_2(9) = \frac{1}{4.1} (4.6 - 1.1x_1(9) - 1.1x_3(8))$$

$$x_2(9) = \frac{1}{4.1} (4.6 - 1.1(0.7382) - 1.1(0.7555))$$

$$x_2(9) = -0.7392$$

$$x_3(9) = \frac{1}{6.2} (5.2 - 2.1x_1(9) - 1.4x_2(9))$$

$$x_3(9) = \frac{1}{6.2} (5.2 - 2.1(0.7382) - 1.4(-0.7392))$$

$$x_3(9) = 0.7556$$

10th iteration when $r=10$

$$x_1(9) = 0.7382, x_2(9) = -0.7392, x_3(9) = 0.7556$$

$$x_1(10) = \frac{1}{4.7} (1.3 - 1.3x_2(9) + 1.6x_3(9))$$

$$x_1(10) = \frac{1}{4.7} (1.3 - 1.3(-0.7392) + 1.6(0.7556))$$

$$x_1(10) = 0.7383$$

$$x_2(10) = \frac{1}{4.1} (4.6 - 1.1x_1(10) - 1.1x_3(9))$$

$$x_2(10) = \frac{1}{4.1} (4.6 - 1.1(0.7383) - 1.1(0.7556))$$

$$x_2(10) = -0.7392$$

$$x_2(0) = \frac{1}{6.2} (5.2 - 1.4(0) - 1.472(0))$$

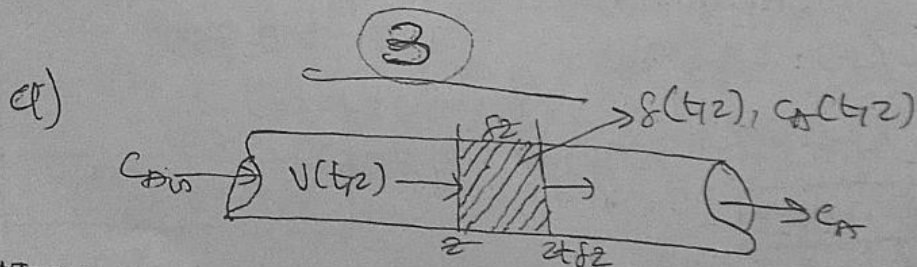
$$x_3(0) = \frac{1}{6.2} (5.2 - 2.1(0.7383) - 1.46(0.7392))$$

$$x_3(0) = 0.7556$$

	0	1	2	3	4	5	6	7
1	1	0.8404	0.7939	0.7136	0.7461	0.7356	0.7291	0.7380
2	1	-0.7706	-0.6876	-0.7834	-0.7342	-0.7408	-0.7386	-0.7393
3	1	0.8974	0.7251	0.7671	0.7518	0.7568	0.7552	0.7557

8	9	10
0.7383	0.7382	0.7383
-0.7391	-0.7392	-0.7392
0.7555	0.7556	0.7556

$$\therefore X = \begin{pmatrix} 0.7383 \\ -0.7392 \\ 0.7556 \end{pmatrix}$$



MODEL FORMULATION

To model a PFR, beginning with performing a macroscopic component balance for species A around the differential element of width δz and constant cross-sectional area S . Hence modelling the plug flow reactor begins in

